



ABF-ZV3

3-way zone valves with electric actuator

1" and 1¼" · Kvs 8 and 13 m³/h · 230 V · 5 N·m



termojet.com.ua

WARNING! Read this manual before starting installation. Electrical connection must be carried out by a qualified electrician only.

Application

The Termojet ABF-ZV3 3-way zone valve is a ready-to-install assembly with a built-in electric actuator for zone control in water supply, heating, ventilation and air-conditioning systems, in heat-pump systems and in water leak protection systems.

Control is on/off: when voltage is applied, the actuator turns the shutter through 60° in 8 seconds and switches the flow between the outlets. There are no intermediate positions. An indicator on the housing shows the current state, and the manual override lets the valve be switched with no power applied.

Technical data

Valve

Maximum working pressure	PN 10
Maximum working temperature	120 °C
Body material	CW617N brass
Shutter	PPS
Size, G	1" and 1¼"
Sealing rings	EPDM
Rubber rings	PA 66 + rubber

Actuator data

Supply voltage	230 V AC or 24 V AC
Control signal	2-point SPST
Operating indicator	yes
Manual override	yes
Protection rating	IP40 (Molex connector), IP44 (cable connection)
Ambient temperature in operation	0...60 °C
Cable	1 m, 3×0.75 mm ²
Running time	8 s per 60°
Direction of rotation	ND, clockwise and counter-clockwise (60°/360°)
Torque	5 N·m
Actuator-to-valve mounting	clip system
Actuator housing material	PA66 + 30 % glass fibre

Range

Part no.	Model	DN	Connection	Kvs, m³/h	Supply
47025230	ABF-ZV3-100	25	1" male	8.0	230 V AC
47032230	ABF-ZV3-114	32	1¼" male	13.0	230 V AC

Both versions share the same 2-point SPST control, 8 s running time per 60°, 5 N·m torque and standard cable. In the factory documentation the same products are designated **ABF01-3-100M** and **ABF01-3-114M**.

Dimensions

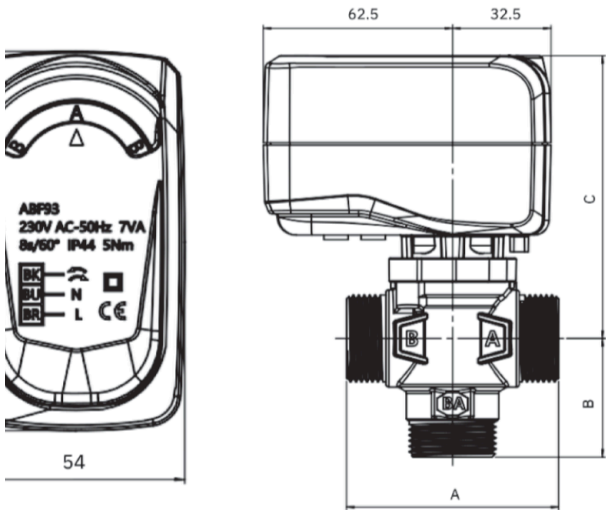


Fig. 1. Overall dimensions, mm

Model	A	B	C
ABF-ZV3-100 (1")	70	39	93
ABF-ZV3-114 (1¼")	88	48	99

Actuator dimensions are the same for both versions: 54 × 95 mm, projection 62.5 and 32.5 mm.

Electrical connection

Connection must be carried out by a qualified electrician only, with the supply line isolated.

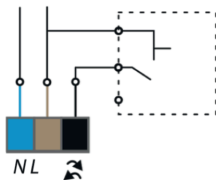


Fig. 2. Wiring diagram, 2-point control

Core colours

Core	Function
 black	control
 blue	neutral, N
 brown	line, L

The **N** and **L** markings are moulded on the actuator itself — use them as the reference when wiring.

Where possible, run the neutral directly from the mains or from the controller. The controller is connected to the supply in accordance with the instructions of its own manufacturer.

Application example

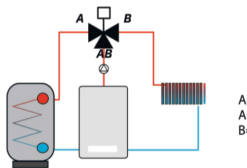


Fig. 3. Switching between the cylinder and the heating circuit — domestic hot water priority

The valve has one inlet and two outlets:

- **AB** — inlet;
- **A** — outlet;
- **B** — outlet.

In a DHW priority layout the water from the boiler enters port AB, and the valve directs it either to the cylinder coil or to the heating circuit, depending on the signal from the thermostat.

Installation

Installation and connection are carried out by qualified trades — a plumber and an electrician. All work is done with the supply voltage switched off.

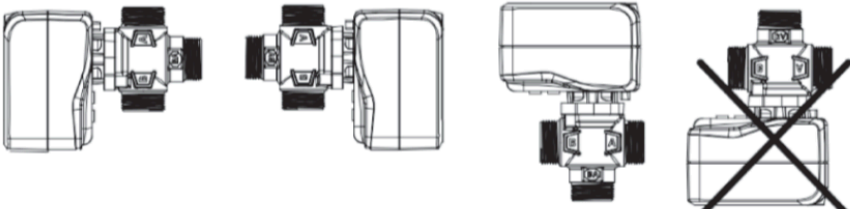


Fig. 4. The actuator fits the valve in one position only. The last variant is not permitted

Note the shape of the valve stem: it has a chamfered cut-out, and the actuator seats on it in one orientation only. Once fitted, the actuator is secured with the split pin.

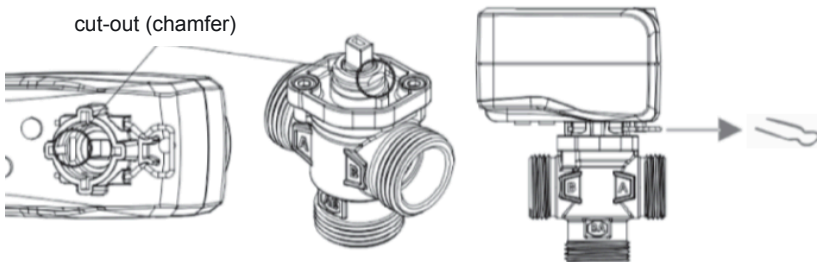


Fig. 5. Cut-out on the stem, actuator seating and split-pin locking

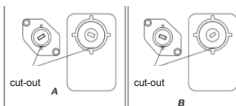


Fig. 6. Two positions of the cut-out — A and B. They determine which outlet is open in the de-energised state

Procedure

1. Shut off the water and isolate the supply line.
2. Fit the valve into the pipework according to the chosen layout; the flow runs from inlet AB to outlets A and B.
3. Determine the required position of the cut-out on the stem — A or B.
4. Seat the actuator on the valve in the only possible position and secure it with the split pin.
5. Connect the cable as shown on p. 4.
6. Switch on the supply and check operation by the indicator on the housing.

Product data

Product	Termojet ABF-ZV3 3-way zone valve
Manufacturer	Termojet
Distributor in Ukraine	Sofiivka-Montazh LLC, EDRPOU 37074476 EDRPOU — Ukrainian company registry code
Address	3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine
Contacts	+380 44 407 24 10 · termojet@sofiievka.kiev.ua termojet.com.ua

Storage and transport

Store the product in the manufacturer's packaging, in heated or unheated warehouses — no closer than one metre to any heating appliance — or under a canopy, in conditions that rule out mechanical damage.

Permissible storage and transport temperature is –20 to +50 °C. The product may be carried by any means of transport in accordance with the carriage rules in force for that mode of transport. During carriage, protect the product from impact, mechanical load and surface scratching.

Disposal

The product contains no substances hazardous to human life and health or to the environment. At the end of its service life, dispose of the product as household waste in accordance with local waste-management rules. Metal parts are suitable for remelting.

Acceptance and testing

The products covered by these instructions have been manufactured, tested and accepted in accordance with the manufacturer's current technical documentation.

Warranty

The manufacturer warrants that the product complies with safety requirements, provided that the user observes the rules set out in these instructions.

The service life of the product, provided that these instructions are observed and the required servicing is carried out, is **10 years** from the date the product is handed over to the user.

The warranty period is **24 months** from the date of sale, but no longer than the service life of the product.

The warranty covers all defects arising through the fault of the manufacturer and **does not cover** defects caused by:

- failure to observe the storage, installation, testing, operation or servicing conditions described here;
- improper transport or loading and unloading;
- exposure to substances aggressive to the materials of the product;
- fire, natural disaster or force majeure;
- acts of the user or third-party interference with the design of the product.

Products that fail within the warranty period because of a manufacturing defect are repaired or replaced free of charge. The cost of removing and transporting the faulty product is not reimbursed to the buyer. If a claim proves to be unfounded, the buyer pays for the diagnosis and examination.

To process a claim the following are required: a free-form statement giving contact details, the installation address and a description of the defect; proof of purchase; photographs of the faulty product, including photographs taken at the place of installation; a copy of the completed warranty card.

Claims are accepted at: 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine. Phone +380 44 407 24 10, e-mail termojet@sofievka.kiev.ua.

The manufacturer reserves the right to introduce changes to the design of the «Termojet ABF-ZV3 3-way zone valve» product that do not impair its quality.

Warranty card

to delivery note No. _____ dated _____ 20____

Product: Termojet ABF-ZV3 3-way zone valve

No.	Part number	Quantity	Notes
1			
2			
3			
4			

Warranty period – 24 months from the date of sale.

I have read and accept the warranty terms and the installation and operating instructions:

Buyer (signature)

Seller (signature)

Date of sale

Stamp or seal
of the selling organisation